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In [22]: import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_csv("C:/Users/SADIYA/Downloads/archive/StudentsPerformance.csv")
df.head()
df.info()
df.describe()
df[['math score','reading score','writing score']].mean().plot(kind='bar')
plt.title("average scores by subjects")
plt.xlabel('subjects')
plt.ylabel('average scores')
plt.show()
print('average scores are slightly higher in other subjects compared to math')
df.groupby('gender')[['math score','reading score','writing score']].mean().plot
plt.title('average scores by gender')
plt.xlabel('gender')
plt.ylabel('average scores')
plt.show()
print("female students performed better in reading and writing where as male stu
plt.hist(df['math score'])
plt.title ('distribution of math scores')
plt.xlabel('math scores')
plt.ylabel('students')
plt.show()
print('most scores scored within middle range with 60-70 being the highest and v
plt.scatter(df['reading score'],df['writing score'])
plt.title('reading score vs writing score')
plt.xlabel('reading score')
plt.ylabel('writing score')
plt.show()
print("Reading and writing scores showed a strong positive relationship, indicat
df.groupby('test preparation course')[['math score', 'reading score', 'writing s
plt.title('Scores Based on Test Preparation Course')
plt.xlabel('Test Preparation Course')
plt.ylabel('Average Score')
plt.show()
print("Students who completed the test preparation course achieved higher averag
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<class 'pandas.core.frame.DataFrame'>
```

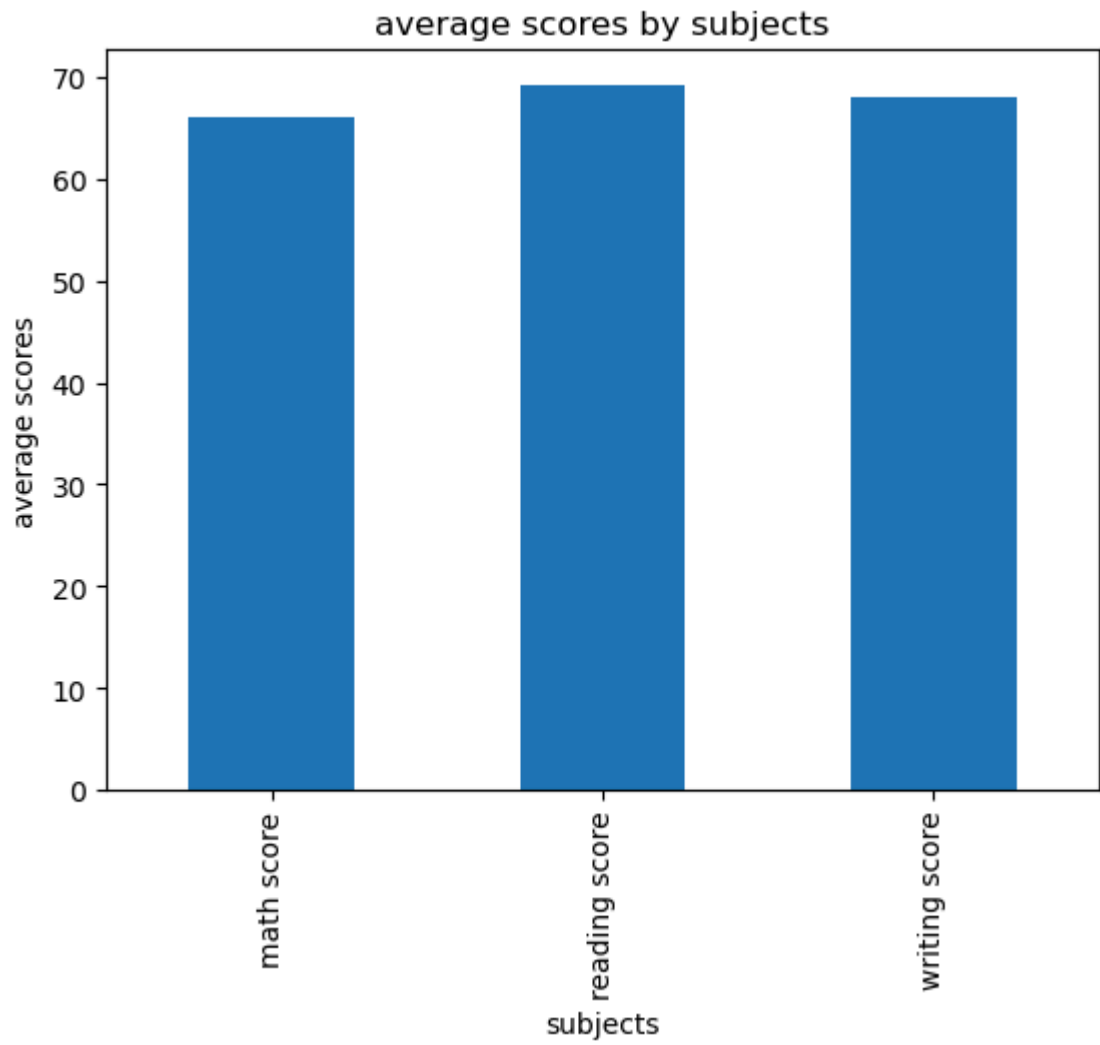
```
RangeIndex: 1000 entries, 0 to 999
```

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Data columns (total 8 columns):
```

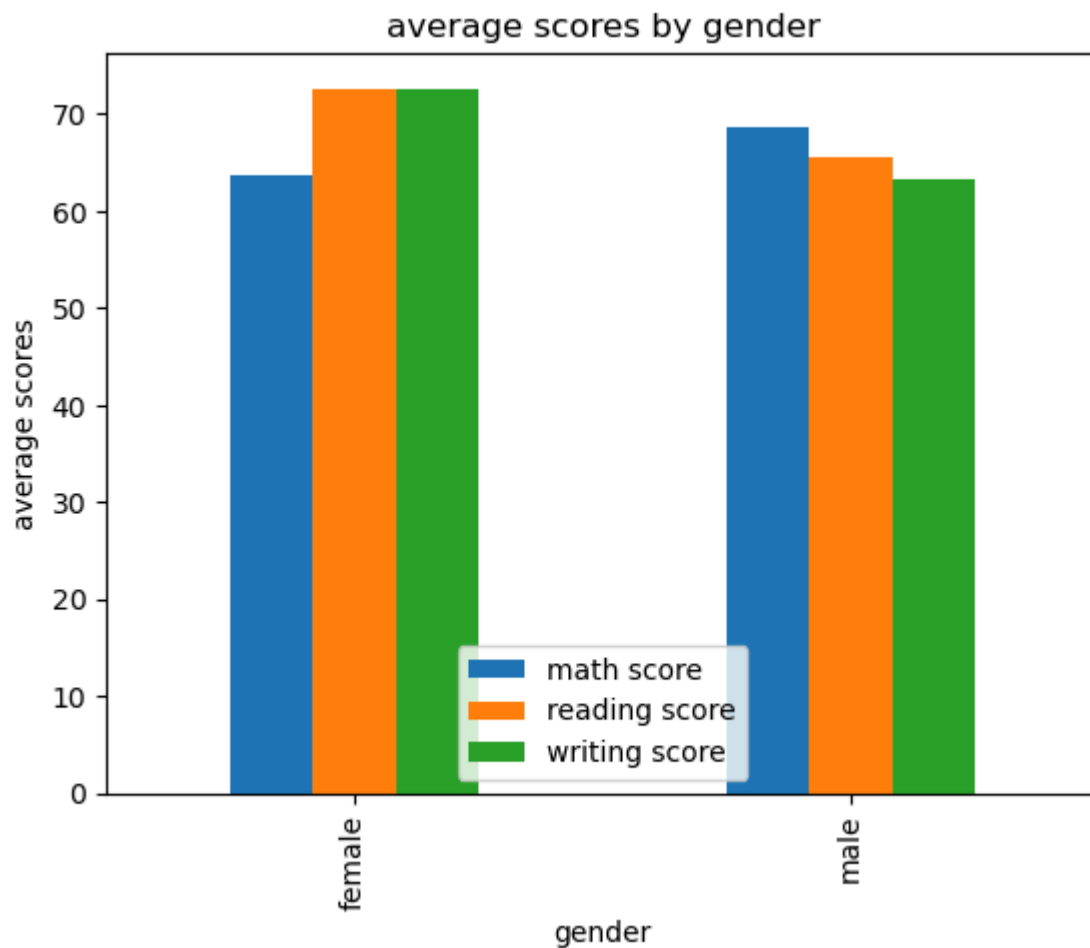
#	Column	Non-Null Count	Dtype
0	gender	1000 non-null	object
1	race/ethnicity	1000 non-null	object
2	parental level of education	1000 non-null	object
3	lunch	1000 non-null	object
4	test preparation course	1000 non-null	object
5	math score	1000 non-null	int64
6	reading score	1000 non-null	int64
7	writing score	1000 non-null	int64

```
dtypes: int64(3), object(5)
```

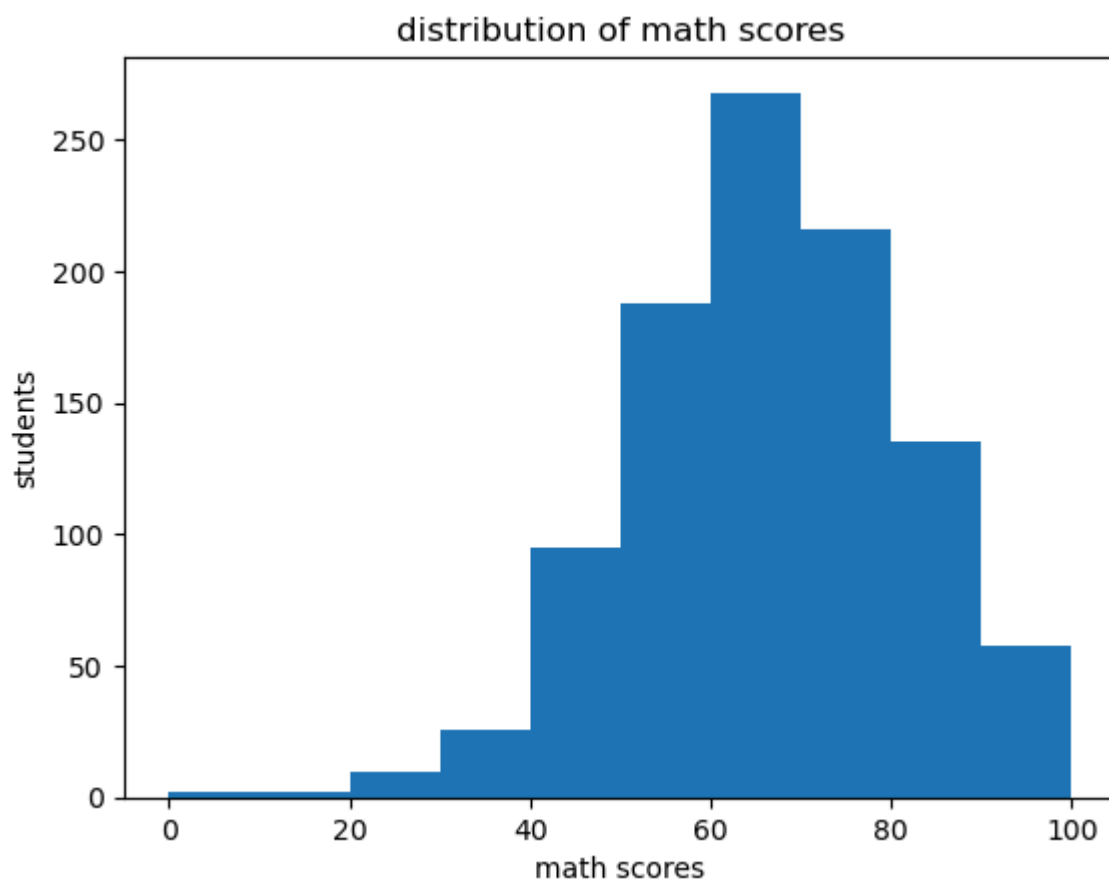
```
memory usage: 62.6+ KB
```



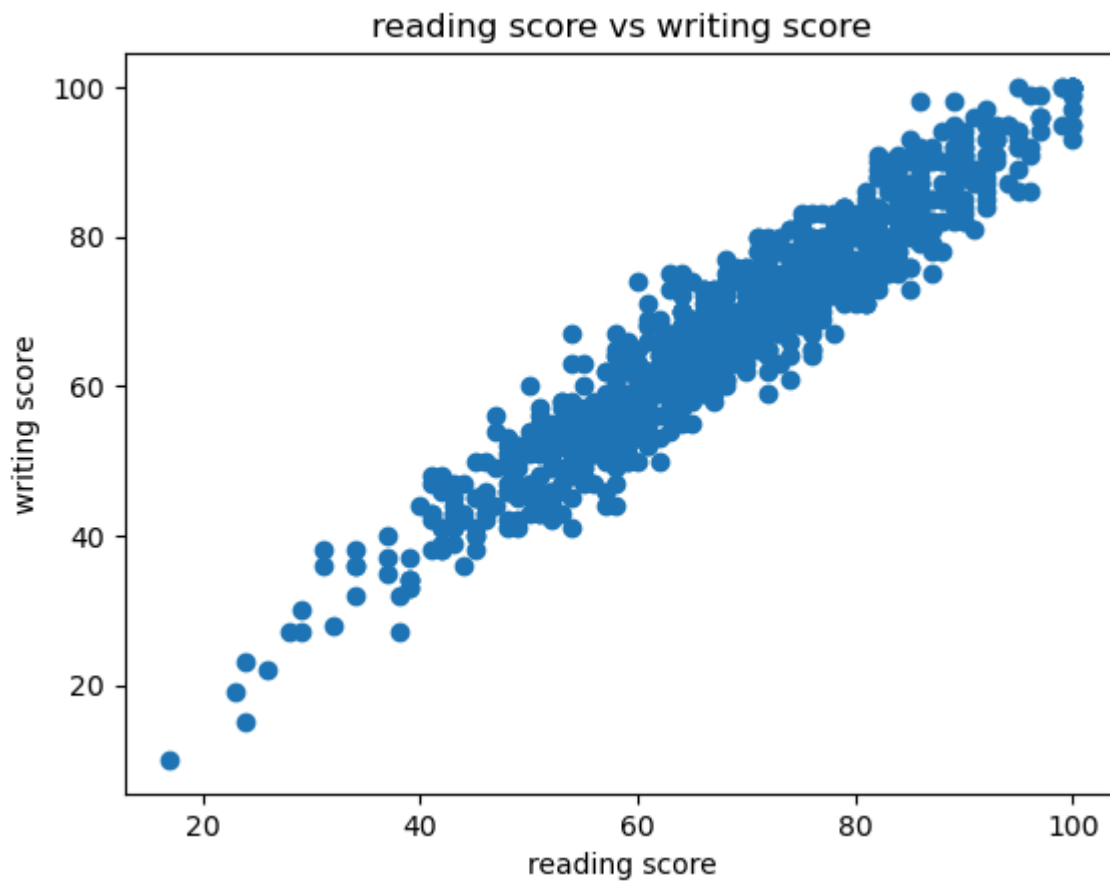
average scores are slightly higher in other subjects compared to math



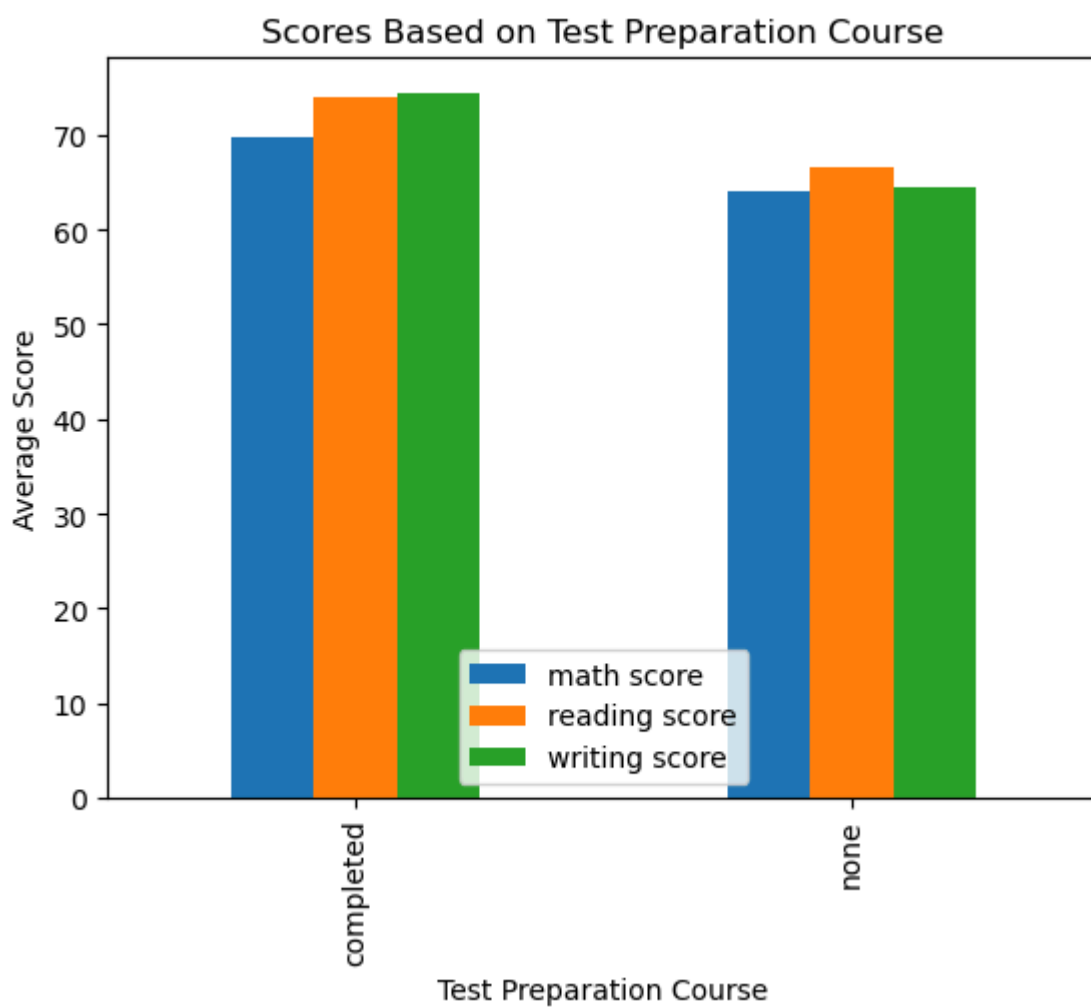
female students performed better in reading and writing where as male students performed better in math



most scores scored within middle range with 60-70 being the highest and very low in the beginning



Reading and writing scores showed a strong positive relationship, indicating that students who performed well in reading also tended to score higher in writing



Students who completed the test preparation course achieved higher average scores across all subjects

In []: